Model Training Report

Introduction

Briefly explain that multiple deep learning models were trained using the preprocessed HAM10000 dataset to classify seven skin disease classes. The objective was to evaluate different CNN architectures and improve their performance through optimization techniques.

Parameter	Value
Dataset	HAM10000
Image Size	224 × 224
Batch Size	32
Framework	TensorFlow / Keras
Loss Function	Categorical Crossentropy
Evaluation Metric	Accuracy

Basic CNN

Model Overview

The Basic CNN model consists of convolutional, pooling, and fully connected layers designed to learn fundamental image features. It serves as the baseline model for evaluating the performance of more advanced architectures.

Deep CNN

Model Overview

The Deep CNN model extends the basic architecture by adding additional convolutional layers, enabling the extraction of more complex and hierarchical image features for improved classification performance.

Batch Normalization

Model Overview

This model incorporates Batch Normalization layers after convolutional layers to stabilize training, accelerate convergence, and improve overall model performance.

CNN Dropout

Model Overview

The CNN + Dropout model applies dropout layers during training to reduce overfitting by randomly deactivating neurons, improving the model's ability to generalize to unseen data.

MobileNetV2

Model Overview

MobileNetV2 is a lightweight transfer learning model that uses depthwise separable convolutions and inverted residual blocks. It provides high classification accuracy with fewer parameters and faster computation.

EfficientNetB0

Model Overview

EfficientNetB0 is a transfer learning model that balances network depth, width, and resolution using compound scaling. It achieves strong performance while maintaining computational efficiency.

DenseNet121

Model Overview

DenseNet121 connects each layer to every subsequent layer, promoting feature reuse and efficient gradient flow. This architecture improves learning efficiency and classification performance while reducing the number of parameters.

Model	Train Accurancy	Validation Accurancy	Test Accurancy	Train Loss	Validation Loss	Test loss
Basic CNN	49.93	49.93	49.93	49.93	49.93	49.93
Deep CNN	47.97	47.97	47.97	47.97	47.97	47.97
Batch Normalization	47.55	47.55	47.55	47.55	47.55	47.55
CNN DropOut	53.24	53.24	53.24	53.24	53.24	53.24
MobileNetV2	60.31	60.31	60.31	60.31	60.31	60.31
EfficientNetB0	55.93	55.93	55.93	55.93	55.93	55.93
ResNet50	59.43	59.43	59.43	59.43	59.43	59.43
DenseNet 121	53.41	53.41	53.41	53.41	53.41	53.41

Observation: Among all the trained models, **MobileNetV2** achieved the highest training, validation, and test accuracy, making it the best-performing baseline model

Model Improvement

Several optimization techniques were applied to improve the performance of the baseline models. These techniques enhanced model accuracy, reduced overfitting, and improved generalization.

Improvement Technique	**Purpose**
Data Augmentation	Increase training data diversity
Batch Normalization	Stabilize model training
Dropout	Reduce overfitting
Early Stopping	Prevent over-training
Learning Rate Scheduling	Improve convergence
Optimizer Comparison	Compare SGD, Adam, and RMSprop
Fine-Tuning	Enhance transfer learning models

Result

The performance of the models improved after applying the optimization techniques. Among the transfer learning models, **ResNet50 trained with the SGD optimizer** achieved the highest accuracy and overall performance. Therefore, the optimized **ResNet50 (SGD)** model was selected as the final model for evaluation, Explainable AI analysis, and deployment in the Flask web application.